4(a)(i)	**M1** (molecules with the) same molecular formula / same number of atoms of each element **M2** different structural formulae/structures	**2**
4(b)(i)	KCN in ethanol (and heat)	**1**
4(b)(ii)	$(C_2H_5)_2CHCN + 2H_2O + HCl \rightarrow (C_2H_5)_2CHCOOH + NH_4Cl$	**1**

Examiner report missing
9701_w25_qp_24 Question 4

4(a)	$C_5H_{10}O_2$	**1**

4(d)(i)	$CH_3CN + HCl + 2H_2O \rightarrow CH_3COOH + NH_4Cl$	**1**

Examiner report missing
9701_s25_qp_24 Question 4

6(a)(i)	two or more reactants / molecules react to make (only) one product	**1**
6(c)(i)	KCN/potassium cyanide in ethanol **AND** heat	**1**
6(c)(ii)	(acid) hydrolysis	**1**
6(c)(iii)	$HOCH_2CH_2CN + 2H_2O + HCl \rightarrow HOCH_2CH_2COOH + NH_4Cl$	**1**

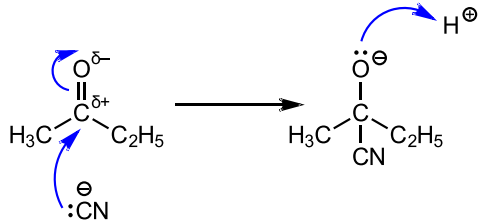
Examiner report missing
9701_s25_qp_23 Question 6

4(a)(i)	$LiAlH_4$	**1**
4(b)(i)	substitution	**1**
4(b)(ii)	KCN / potassium cyanide in ethanol / C_2H_5OH **AND** heat	**1**
4(b)(iii)	$CH_3CH_2CH_2CN + NaOH + H_2O \rightarrow CH_3CH_2CH_2CO_2Na + NH_3$ **OR** $CH_3CH_2CH_2CN + NaOH + 2H_2O \rightarrow CH_3CH_2CH_2CO_2Na + NH_4OH$ **M1** correct organic product (as molecule or anion) **M2** rest of equation is correct	**2**
4(b)(iv)	(alkaline) hydrolysis	**1**

Examiner report missing
9701_s25_qp_22 Question 4

4(a)		**5**

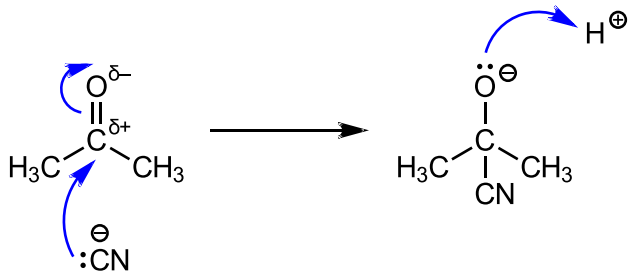
acidified $K_2Cr_2O_7$(aq)	(solution) turns (from orange to) green	**J** and **L**
alkaline I_2(aq)	yellow precipitate forms	**K** and **L**
2,4-dinitrophenylhydrazine / 2,4-DNPH	orange precipitate forms	**J** and **K**
Na(s)	effervescence	**L** and **M**

4(b)(i)		**4**

- curly arrow from lone pair on :CN to C($^{\delta+}$)
- curly arrow from C=O to O($^{\delta-}$) **AND** correct dipole
- correct intermediate
- curly arrow from lone pair on O:$^-$ to H^+

4(b)(ii)	hydroxynitrile	**1**

Examiner report missing
9701_m25_qp_22 Question 4

4(a)(i)	 • curly arrow from lone pair on C of CN^- to carbonyl C • correct dipole on C=O • curly arrow from C=O bond to O • correct intermediate (the lone pair(s) not needed for this bullet point) • curly arrow from lone pair on —O to (source of) H^+	**1**
		1
		1

4(b)	H_2SO_4(aq) or HC*l*(aq)	**1**

Question 4

(a) (i) The mechanism attracted many good answers. The common errors included: arrows from the negative charge (on the CN^- ion) instead of from the lone pair on the C atom, missing polarity, specifically missing the $\delta+$ on the H (of H-CN), and incorrect attack of the intermediate by the CN^- ion.

(ii) Many answers stated that there were two CH_3 groups on one C atom in A but did not go further to explain the significance of this in terms of the lack of optical isomerism in A. Credit was given for stating that no chiral centre was present or that the central C atom was not bonded to four different atoms/groups.

(b) Candidates often stated that the mineral acids HCl or H_2SO_4 were required for the hydrolysis reaction of a CN group to a carboxylic acid but missed the requirement for the acid to be dilute or aqueous.

(c) (i) The equation was frequently correct. A common error was the omission of 2(H), from the equation, or using just 2H, without the brackets.

(ii) The name of compound C was well known.

(d) Most candidate answers stated that an orange precipitate was formed when propanone was reacted with 2,4-DNPH.

(e) A common answer was that 'Fehling's reagent only reacts with aldehydes', with no reference to any chemistry in this reaction. Fehling's reagent behaves as an oxidising reagent and ketones are not easily oxidised, hence, there is no reaction between Fehling's reagent and propanone.

(f) (i) Many candidate answers stated that the absorptions at 2850 – 2950 cm^{-1} were attributable to a C-H bond. As all three compounds in the question contained groups containing C-H bonds these compounds were indistinguishable using these absorptions.

(ii) Many candidates correctly identified the absorption at 2200 – 2250cm^{-1}, which was unique to compound A, as it was the only compound containing a C≡N bond. As all three compounds, A, B and C contained an O-H bond, no credit was given for frequent mention of this absorption.

2(a)(i)	M1 $4Na + O_2 \rightarrow 2Na_2O$	**1**
	M2 $S + O_2 \rightarrow SO_2$	**1**

2(b)(ii)	$CH_3CN + 2H_2O + (1)H^+ \rightarrow CH_3COOH + NH_4^+$	**1**

Question 2

(a) (i) The formula of sodium oxide was not well known. In equations where correct formulae were shown, a significant proportion were not balanced.

(ii) Candidate answers were rarely given in terms of an aluminium cation and an oxide anion. When ions were described, they were generally shown with the appropriate number of outer electrons; charges on the ions were sometimes omitted or incorrect.

(iii) Weak candidate answers did not fully address the question and explained why the maximum oxidation state varies because of the variable number of valence electrons.

(b) (i) Many candidate answers referred to $Mg(OH)_2$ with a pH of a slightly acidic solution rather than weakly alkaline. The product of the reaction of P_4O_{10} and H_2O was not well known, $P(OH)_5$ was a common incorrect answer.

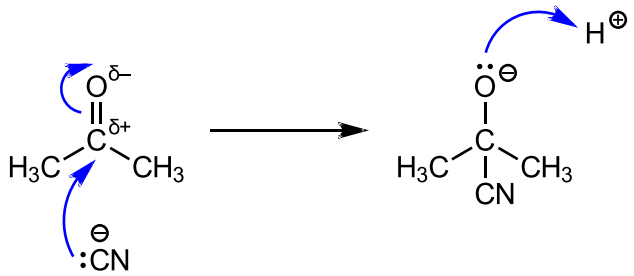
(ii) This equation proved to be demanding. Common errors included equations which were incorrectly balanced or showed ammonia as a product of the reaction.

(iii) This question was answered well by candidates. Occasionally the same structure was shown twice, usually with propan-1-ol as both answers.

(iv) Excellent candidate answers were seen and these explained why there were fewer $H^+(aq)$ in solution in terms of the increase in strength of O-H bond due to the positive inductive effect of the ethyl group. Inappropriate use of the term 'electronegativity' instead of 'electron density' was seen. Common misconceptions included the production of OH^- ions when alcohols dissociate or that water fully dissociates or that alcohols contain OH^- ions.

(c) (i) Many candidate answers referred to H_2O although this was not relevant in **(c)(i)**. Weaker responses appreciated that there was an increase in strength of intermolecular forces but did not explain how this change arose. Many candidates gave answers in terms of an increase in size of the molecules or in terms of nuclear attraction, nuclear charge and shielding, which is incorrect. A common misconception described the increase in boiling point due to an increase in strength of permanent dipoles as the group is descended.

(ii) Identification of hydrogen bonds in H_2O was relatively common. The relevant strength of hydrogen bonds in water compared to those intermolecular forces present in the other hydrides was not always included. Incorrect reference to H_2S containing hydrogen bonds was seen and there was ambiguity in answers that referred to the relative strength of the O-H bonds rather than intermolecular hydrogen bonds.

4(a)(i)	 • curly arrow from lone pair on C of CN^- to carbonyl C • correct dipole on C=O • curly arrow from C=O bond to O • correct intermediate (the lone pair(s) not needed for this bullet point) • curly arrow from lone pair on —O to (source of) H^+	**1**
		1
		1

4(b)	H_2SO_4(aq) or HC*l*(aq)	**1**

Question 4

(a) (i) The mechanism attracted many good answers. The common errors included: arrows from the negative charge (on the CN^- ion) instead of from the lone pair on the C atom, missing polarity, specifically missing the $\delta+$ on the H (of H-CN), and incorrect attack of the intermediate by the CN^- ion.

(ii) Many answers stated that there were two CH_3 groups on one C atom in A but did not go further to explain the significance of this in terms of the lack of optical isomerism in A. Credit was given for stating that no chiral centre was present or that the central C atom was not bonded to four different atoms/groups.

(b) Candidates often stated that the mineral acids HCl or H_2SO_4 were required for the hydrolysis reaction of a CN group to a carboxylic acid but missed the requirement for the acid to be dilute or aqueous.

(c) (i) The equation was frequently correct. A common error was the omission of 2(H), from the equation, or using just 2H, without the brackets.

(ii) The name of compound C was well known.

(d) Most candidate answers stated that an orange precipitate was formed when propanone was reacted with 2,4-DNPH.

(e) A common answer was that 'Fehling's reagent only reacts with aldehydes', with no reference to any chemistry in this reaction. Fehling's reagent behaves as an oxidising reagent and ketones are not easily oxidised, hence, there is no reaction between Fehling's reagent and propanone.

(f) (i) Many candidate answers stated that the absorptions at 2850 – 2950 cm^{-1} were attributable to a C-H bond. As all three compounds in the question contained groups containing C-H bonds these compounds were indistinguishable using these absorptions.

(ii) Many candidates correctly identified the absorption at 2200 – 2250cm^{-1}, which was unique to compound A, as it was the only compound containing a C≡N bond. As all three compounds, A, B and C contained an O-H bond, no credit was given for frequent mention of this absorption.

5(a)	cold dilute $KMnO_4$ **OR** cold dilute potassium manganate(VII)	**1**
5(d)(i)	**W** =	**1**
5(d)(ii)	NH_4Cl	**1**

Question 5

This question tested organic reactions and spectroscopic methods using cyclohexene as a starting material. Many candidates seemed familiar with the chemistry, although the context of a cyclic molecule proved to be disconcerting for a few, who appeared to misinterpret the information given, or did not concentrate solely on the relevant functional groups.

(a) This was well answered by many, although differences appeared in the conditions to use with the $KMnO_4$ reagent (e.g. concentrated/dilute, alkaline/neutral/acidic).

(b)(i) Step 1 was familiar to many candidates. The hydrolysis of two functional groups in step 2 led many to think that H_2O_2 was being used, rather than two portions of NaOH.

(ii) In asking for a mechanism, 'electrophilic' was expected as well as 'addition'.

(c) Candidates showed some understanding in answering this question but often missed full credit by not following the rubric properly: the diagram needed labelling and each responsible bond (stretch) needed to be clearly identified. Many candidates only achieved half of this, or switched **F** and **G**.

(d)(i) This was mostly well answered, although some candidates lost credit when opting to give fully displayed versions of the CN group, neglecting to show a triple C≡N bond or using the wrong connectivity (N attached to the ring).

(ii) This was answered well by many candidates. Others did not acknowledge that acidic conditions are used, creating an ammonium species rather than NH_3.

(iii) The formulae of the organic products were generally well indicated, though candidates often then misbalanced the equation with insufficient (H).

(iv) This was generally well answered by candidates. A few $NaBH_4$ responses were offered, to no credit: carboxylic acids need the stronger of the two hydride reagents on the syllabus.

4(a)	<table><tr><td>molecule</td><td>**P**</td><td>**Q**</td><td>**R**</td><td>**S**</td></tr><tr><td>geometrical</td><td>✓</td><td>(×)</td><td>(×)</td><td>(×)</td></tr><tr><td>optical</td><td>(×)</td><td>✓</td><td>✓</td><td>✓</td></tr></table>	**1**
	<table><tr><td>molecule</td><td>**P**</td><td>**Q**</td><td>**R**</td><td>**S**</td></tr><tr><td>geometrical</td><td>✓</td><td>(×)</td><td>(×)</td><td>(×)</td></tr><tr><td>optical</td><td>(×)</td><td>✓</td><td>✓</td><td>✓</td></tr></table> $CH_3CHBrCH_2CH_3$	**1**
4(b)(iii)	$C_4H_9CN + 2H_2O + H^+ \rightarrow C_4H_9COOH + NH_4^+$	**1**

Question 4

This an organic and spectroscopic question. Candidates are advised to be clear in matters of structure, both when representing molecules diagrammatically and when referring to features within a molecule, such as those identified by different types of spectra.

(a) This was answered well by many candidates. There were incidences of ambiguous answers, such as a 'crossed tick' (✓✗). Credit cannot be awarded if an answer is ambiguous.

(b) (i) Unambiguous structural formulae only were accepted, therefore molecular formulae were discounted and those in which the connectivity was questionable.

(ii) There were many correct answers to this question.

(iii) This proved to be one of the most challenging questions on the paper, requiring the balanced inclusion of (2)H_2O and H^+ on the left-hand side, and NH_4^+ on the right. Salts from viable reactions of specific acids were accepted.

(c) (i) Candidates who understood the $M^+ : [M+1]^+$ ratio were generally able to gain full credit here. Occasional arithmetical errors were seen.

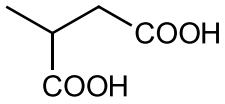
(ii) Many candidates gave 'ester' as a correct answer but were then unable to corroborate this with sufficient and necessary information from the spectrum. Reference to an absorption outside of the C=O ester region contradicted an answer; equally, an absorption within the 1710–1750 cm^{-1} window was not solely sufficient, because of overlap with other absorptions. Corroboration could be given with reference to the C–O stretch (and the lack of O–H stretch).

C–H stretches are always ignored in these situations.

Absorptions must be accompanied by with assignment to an identified bond.

(iii) Propan-1-ol was a common incorrect answer for **T**. OH– is discouraged as a representation of an alcohol group. Many candidates did not draw the branched C5 acid correctly: some structures were unbranched; others had too many or too few H atoms.

2(a)(i)	increases down the group	**1**

2(c)	CH_3OH	**1**
	 one COOH group **AND** correct carbon framework (5C's) (each C has the correct valency)	**1**
	both COOH groups	**1**

Question 2

(a) (i) The solubilities of group 2 hydroxides 'increases' down the group; this variation in solubilities was correctly stated by many candidates

(ii) When barium oxide is added to water the observation is that the white solid dissolves/disappears to form a colourless solution. A common error was that there was bubbling/effervescence seen.

(iii) In the given reaction scheme, reaction 2 involved the formation of $BaCO_3$ and water from adding a reactant to $Ba(OH)_2$. Carbon dioxide, CO_2, is a reactant that would form both the carbonate and water. Many answers included a metal carbonate which would not give water as a product.

(iv) The addition of ethanoic acid to $Ba(OH)_2$ gave barium ethanoate, $(CH_3COO)_2Ba$, as the product.

Many candidates appreciated that barium ethanoate was formed but opted to answer using a formula of $(CH_3COO)Ba$, which was not creditworthy.

(v) $Ba(OH)_2$ and hydrogen are formed when Ba is added to water and the equation is:

$$Ba + 2H_2O \rightarrow Ba(OH)_2 + H_2$$

A common error was to use only 1 mole of water molecules and water formed as a product.

(b) (i) The thermal stability of group 2 carbonates increases down the group; hence, when barytocalcite, $BaCa(CO_3)_2$, is heated $CaCO_3$ will decompose first. Most answers correctly stated that $CaCO_3$ would decompose first but explanations compared the reactivity of Ca with Ba.

(ii) The equation to represent the thermal decomposition of $BaCa(CO_3)_2$ was often correct.

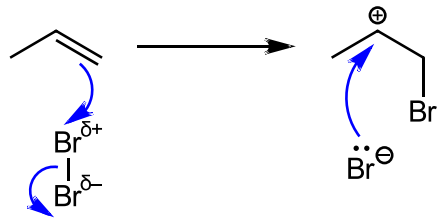
$$BaCa(CO_3)_2 \rightarrow BaO + CaO + 2CO_2$$

Common errors included answers with BaCa as a product or omitting $2CO_2$ as a product.

(c) Full credit for the hydrolysis reaction was rarely awarded. The two organic products are methanol, CH_3OH, and the diacid.

That both the –CN and $–CO_2CH_3$ groups will undergo hydrolysis was not recognised by many candidates. The –CN group was frequently left unhydrolysed and the product methanol was not given as the second organic product.

4(a)(i)	(it has molecules that) contain a chiral carbon / centre **OR** are non-superimposable mirror images	**1**

4(b)(i)	 curly arrow from C=C bond to $Br^{\delta+}$ **AND** curly arrow from $Br^{\delta+}$— $Br^{\delta-}$ bond to $Br^{\delta-}$ **AND** $Br^{\delta+}$ closest to the double bond	**1**
	correct intermediate	**1**
	curly arrow from lone pair on Br^- to $C^{(+)}$ of the intermediate	**1**

4(b)(iii)	HCN&KCN **OR** NaCN with H_2SO_4	**1**
	NaOH(aq) (1) dilute / aq(ueous)HC*l* / H_2SO_4 (1)	**2**
	acidified $K_2Cr_2O_7$	**1**

Question 4

(a) (i) Some candidates gave clear explanations of the presence of a chiral carbon in the lactic acid structure. A few candidates stated that optical isomers had chiral centres or described the behaviour of an optical isomer in the presence of polarised light.

(ii) Many candidates were unable to give the systematic name of lactic acid. 'Carboxylic acid' was seen frequently.

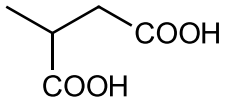
(iii) It was common for diagrams to show **more** than one water molecule or **multiple** hydrogen bonds. Candidates should be advised to respond as directed by the question. It was quite common for answers, showing more than one hydrogen bond, to contradict a correct representation.

(b) (i) Some clear and precise reaction mechanisms were shown with partial charges assigned to relevant atoms and unambiguous representation of curly arrows to show bonds broken and bonds made. When drawing mechanisms of this type candidates should be encouraged to consider what the curly arrows represent and whether their representation is clear, use partial charges correctly and discriminately and check that the number of carbon atoms on the intermediate is correct.

(ii) The equations described sometimes contained only pyruvic acid as a product or identified H_2 rather than H_2O as a product.

(iii) This question was demanding. Many candidates confused the reagents and conditions required to convert halogenoalkanes to nitriles with those required to convert ethanal to a hydroxynitrile. Deduction of the reagents and conditions used in the second step of both synthesis from propene and ethanal proved demanding for most candidates. Oxidising agents were more commonly seen for the third steps, however, some candidates represented the oxidising agents with incorrect formulae or with no reference to acid conditions.

2(a)(i)	increases down the group	**1**

2(c)	CH_3OH	**1**
	 one COOH group **AND** correct carbon framework (5C's) (each C has the correct valency)	**1**
	both COOH groups	**1**

Question 2

(a) (i) The solubilities of group 2 hydroxides 'increases' down the group; this variation in solubilities was correctly stated by many candidates

(ii) When barium oxide is added to water the observation is that the white solid dissolves/disappears to form a colourless solution. A common error was that there was bubbling/effervescence seen.

(iii) In the given reaction scheme, reaction 2 involved the formation of $BaCO_3$ and water from adding a reactant to $Ba(OH)_2$. Carbon dioxide, CO_2, is a reactant that would form both the carbonate and water. Many answers included a metal carbonate which would not give water as a product.

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Many candidates appreciated that barium ethanoate was formed but opted to answer using a formula of $(CH_3COO)Ba$, which was not creditworthy.

(v) $Ba(OH)_2$ and hydrogen are formed when Ba is added to water and the equation is:

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A common error was to use only 1 mole of water molecules and water formed as a product.

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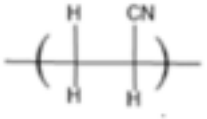
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$$BaCa(CO_3)_2 \rightarrow BaO + CaO + 2CO_2$$

Common errors included answers with BaCa as a product or omitting $2CO_2$ as a product.

(c) Full credit for the hydrolysis reaction was rarely awarded. The two organic products are methanol, CH_3OH, and the diacid.

That both the –CN and $–CO_2CH_3$ groups will undergo hydrolysis was not recognised by many candidates. The –CN group was frequently left unhydrolysed and the product methanol was not given as the second organic product.

6(a)	 OR 	**1**
6(f)(i)	substitution	**1**
6(f)(ii)	**M1** ammonia / NH_3 **M2** heat + (M1 in ethanol) under pressure	**2**

Question 6

(a) The single repeat unit derived from the monomer, CH_2=CHCN, by addition polymerisation was quite well answered.

(b) (i) Many candidates recognised that the mechanism for step 1 was nucleophilic addition.

(ii) Overall, the mechanism for step 1 was generally not answered well. The polarity on the C=O bond was often absent and the arrow from the CN^- to the C (in the C=O bond) did not start from the lone pair on the C of the CN^- ion. Several answers contained an incorrect intermediate with a positive charge which was then attacked by the CN^- nucleophile and many answers incorrectly showed an arrow from an H^+ ion to an O^- (in the intermediate).

(iii) Some good answers were seen.

(iv) The absorptions in the infrared spectrum of CH_2=CHCN were often correctly identified as C≡N and C=C. Incorrect answers mainly contained an identification of the functional groups rather than the bond or identifying the absorption (annotated as T on the IR spectrum) as that of a C=O group.

(c) (i) Very few candidates gave an accurate description.

(ii) A number of answers were awarded some credit. Few candidates were able to give a full description.

(d) Several candidates gave a correct pair of molecules that were non-superimposable images of $CH_3CH(OH)CN$. A common error was diagrams containing the same type of bond at 180° to each other.

(e) The equation for the reduction of propanenitrile to propylamine using hydrogen gas was not well known. Few candidates used the correct stoichiometry for hydrogen.

(f) (i) It was well known that the name of the type of reaction in step 1 is substitution.

(ii) Many candidates gained credit for ammonia. Many omitted heat or the solvent as conditions.

3(a)(i)	KCN in ethanol + heat	**1**

3(a)(iii)	(acid) hydrolysis	**1**

Question 3

(a) (i) Some candidates incorrectly described the reagent and conditions required to convert carbonyl compounds to hydroxynitriles, rather than halogenoalkanes to nitriles.

(ii) The systematic name for **X** was not always used, e.g. propionitrile or ethyl cyanide. Omission of the 'e' in propanenitrile was often seen.

(iii) It was generally well known that this type of reaction is a hydrolysis reaction.

(iv) Many candidates correctly produced equations representing the production of hydrogen gas. The formula of the calcium salt was frequently given in terms of calcium with a valency of one.

(b) Many candidates knew that acidic conditions were required. Stronger candidates appreciated that the essential condition for this reaction was 'heat under reflux' to ensure complete oxidation to the carboxylic acid.

(c) Most candidates described the correct number of pi bonds present in **X**; the correct number of sigma bonds was seen less frequently. Seven, nine and ten sigma bonds were common incorrect answers.